Royal University of Phnom Penh
Faculty of Engineering
Major: Data Science and Engineering

EXPLAINABLE AI FOR DIABETES RISK PREDICTION AND ACTIONABLE INSIGHTS

Presenter: Nhor Povketya
Advisor: Dr. Khim Chamroeun

OUTLINE

- Background and Motivation
- Research Objective
- Literature Review
- Methodology
- Results
- Conclusion
- Demo

BACKGROUND AND MOTIVATION

- Diabetes is a chronic disease associated with serious complications such as heart disease, kidney failure, and vision loss
- Early risk prediction can support prevention and screening
- Machine learning identifies risk patterns from health and lifestyle survey data
- Explainable AI helps users understand model predictions, reveals reasoning behind model predictions

Introduction

PROBLEM STATEMENT

01 Traditional diabetes screening may require time, cost, and medical resources

02 Existing prediction systems often focus only on prediction accuracy without explaining the results clearly

03 Cambodia has limited AI-based diabetes prediction research using local health data

04 Health datasets may contain missing values and imbalanced data, which can affect model performance

05 Need for explainable risk prediction system to raise awareness and support early screening

RESEARCH OBJECTIVES

- Develop machine learning models for diabetes risk prediction using survey-based health indicators
- Train and validate the models using BRFSS 2017 data, and evaluate the final model on Cambodia STEPS as an independent test set
- Compare model performance using Recall, Precision, Accuracy, and F1-score
- Apply Explainable AI (SHAP) to identify key diabetes risk factors and explain individual predictions
- Generate actionable insights to support diabetes risk awareness and early screening

RESEARCH QUESTIONS



RQ1

Can health survey data from Cambodia STEPS and BRFSS be used to predict diabetes risk without blood test data?



RQ2

Which machine learning model is the best for finding people who may have diabetes risk?



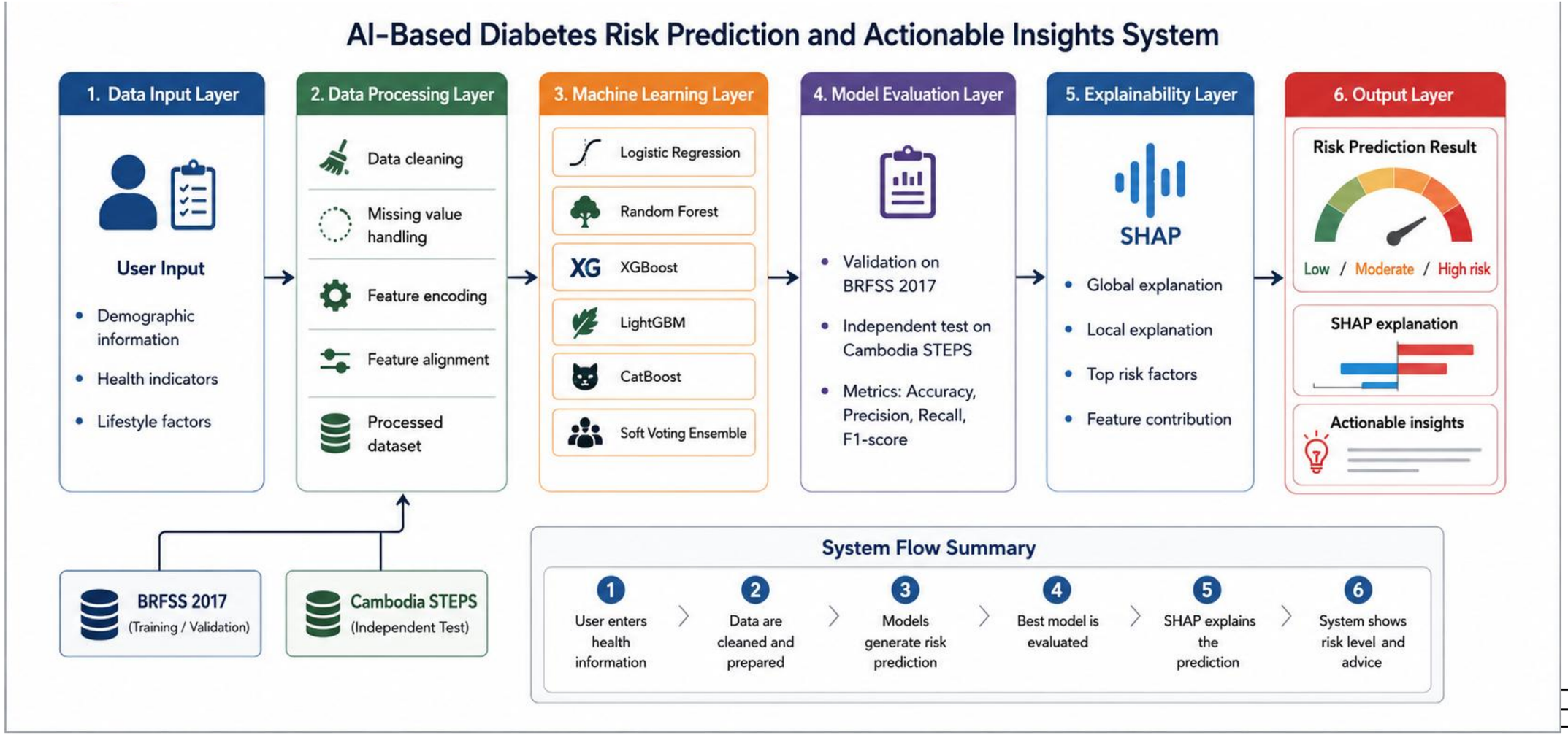
RQ3

Which health indicators contribute most to diabetes risk prediction according to SHAP analysis?

LITERATURE REVIEW

Paper / Author	Model Used	Dataset Type	Accuracy / Result
Xie et al. (2019) — *Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques*	SVM, Decision Tree, Logistic Regression, Random Forest, Neural Network, Naive Bayes	BRFSS 2014 survey data	Neural Network: **82.4% accuracy**, AUC **0.7949**. Decision Tree had highest sensitivity **51.6%**
Khokhar et al. — *Towards Transparent and Accurate Diabetes Prediction Using ML and XAI*	Random Forest, XGBoost, LightGBM, Ensemble Model + SHAP/LIME	Diabetes Binary Health Indicators dataset from BRFSS 2015	Ensemble model: **92.50% accuracy**, ROC-AUC **0.975**. Important features: BMI, Age, General Health, Income, Physical Activity
Katiyar et al. (2024) *Recent Advancements Using ML & DL Approaches for Diabetes Detection*	ML and Deep Learning Models	Multiple Diabetes Datasets	ML and DL models achieved high prediction performance, dataset quality significantly affects results
Li et al. (2025) Performance of Diabetes Risk Prediction Models	Statistical Models + Machine Learning Models	Population Health Datasets	Risk prediction models can support early screening and prevention of Type 2 Diabetes
Khokhar et al. (2025) Advances in Artificial Intelligence for Diabetes Prediction	Logistic Regression, SVM, Random Forest, XGBoost, CNN, RNN	Multiple Public Diabetes Datasets	Reviewed 53 studies, highlighted strong performance of ML models and the need for explainable and clinically relevant systems

System Architecture



DATASET OVERVIEW

Cambodia STEPS

Local survey data collected in Cambodia

- Demographic indicators (age, sex, education)
- Health indicators (BMI, blood pressure, cholesterol)
- Lifestyle indicators (smoking, physical activity, diet)
- 873 rows after preprocessing with 14% of Positive rate
- Target: Diabetes status (0 = non-diabetes, 1 = diabetes)

BRFSS

Larger benchmark dataset from the United States

- Similar health and lifestyle indicators
- Used for training 2017
- 275746 rows after preprocessing 16% of Positive rate
- Target: Diabetes status (0 = non-diabetes, 1 = diabetes)

SELECTED FEATURES



Demographic

- Age
- Sex
- Education



Health Status

- High Blood Pressure
- High Cholesterol
- BMI



Lifestyle & Healthcare Access

- Smoker
- Physical Activity
- Fruits
- Vegetables
- Heavy Alcohol Consumption

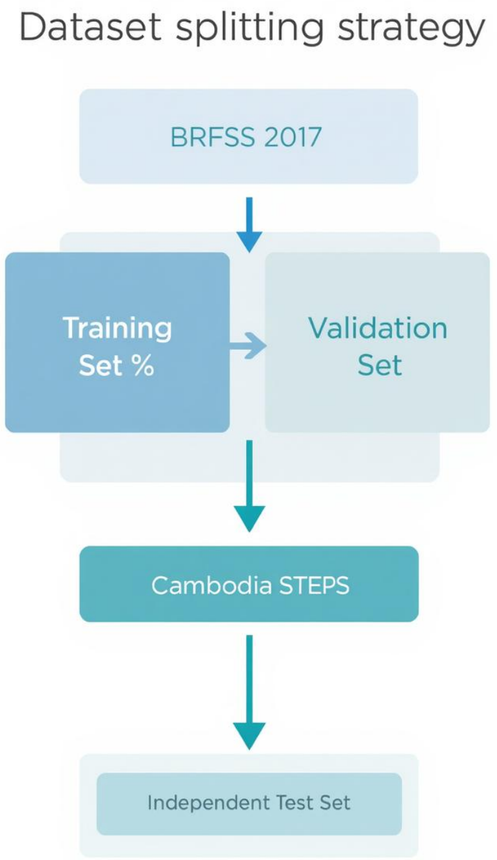
DATA SPLITTING STRATEGY

BRFSS 2017 Dataset

- 80% Training Set: Used for model learning and fitting
- 20% Validation Set: Used for model selection and decision threshold tuning

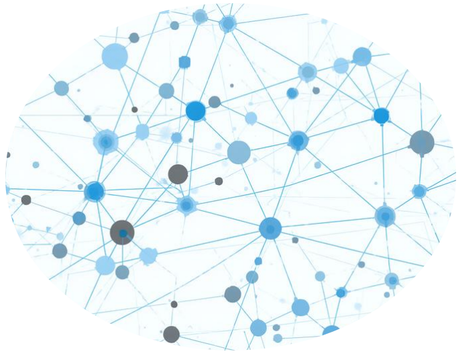
Cambodia STEPS Dataset

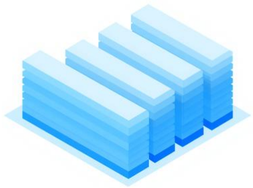
- Independent Test Set: Used for final cross-population evaluation only



MACHINE LEARNING MODELS

1. Logistic Regression
2. Random Forest
3. XGBoost
4. LightGBM
5. CatBoost
6. Soft Voting Ensemble





EVALUATION METRICS

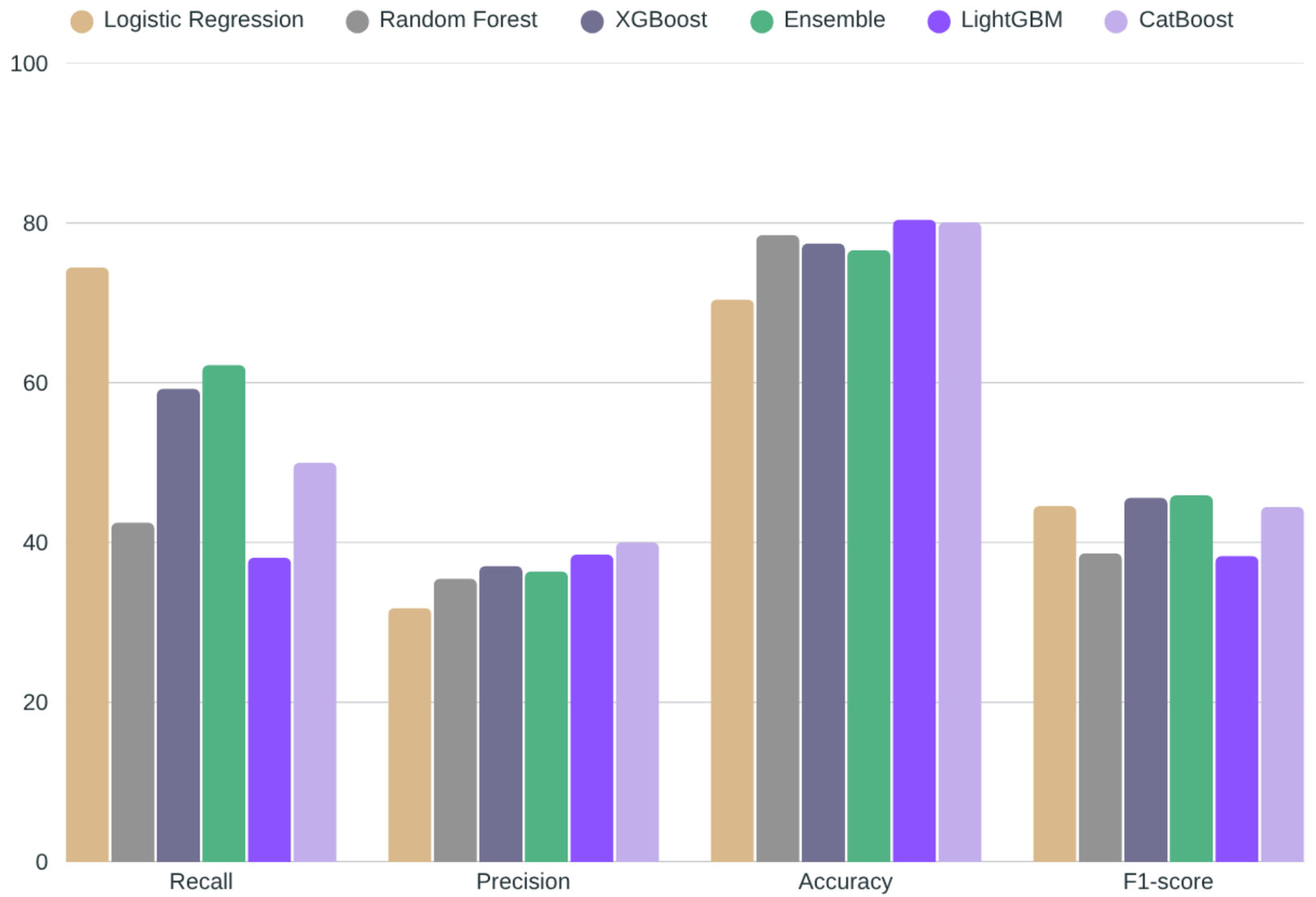
Recall

Precision

Accuracy

F1 Score

Model Evaluation and Results

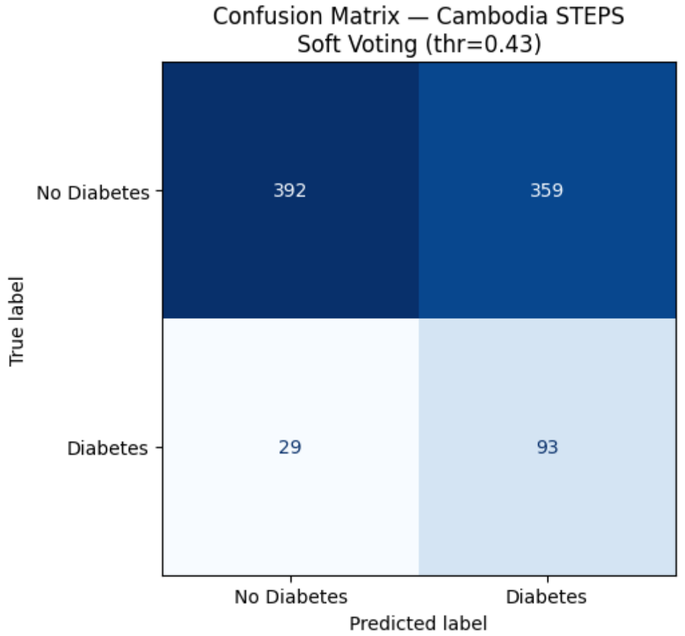


Confusion Metrices



CAMBODIA STEPS INDEPENDENT TEST RESULT

- Final selected model: Soft Voting Ensemble
- Lower performance is expected because the datasets come from different populations and survey systems




```
============================================================
  INDEPENDENT TEST — Cambodia STEPS (once)
============================================================
  Model     : Soft Voting
  Threshold : 0.43
  Train data: BRFSS 2017 only
  Features  : 11 (['Sex', 'Age', 'Education', 'HighBP',

  Recall       0.7623 <- clinical priority
  Precision    0.2058
  Accuracy     0.5556
  F1           0.3240 <- selection metric

              precision    recall  f1-score   support

 No Diabetes       0.93      0.52      0.67       751
    Diabetes       0.21      0.76      0.32       122

    accuracy                           0.56       873
   macro avg       0.57      0.64      0.50       873
weighted avg       0.83      0.56      0.62       873
```

	Validation (BRFSS 2017)	Test (STEPS)
Recall	0.7046	0.7623
F1	0.4515	0.3240
Accuracy	0.7269	0.5556
Precision	0.3322	0.2058

EXPLAINABLE AI

Global Explanation

Overall feature importance across the entire dataset using SHAP summary plots.

- Identifies which features most influence diabetes risk predictions globally
- BMI, age, blood pressure, and general health rank as top contributors
- SHAP summary plot visualizes direction and magnitude of each feature's impact
- Enables model-level transparency for researchers and practitioners

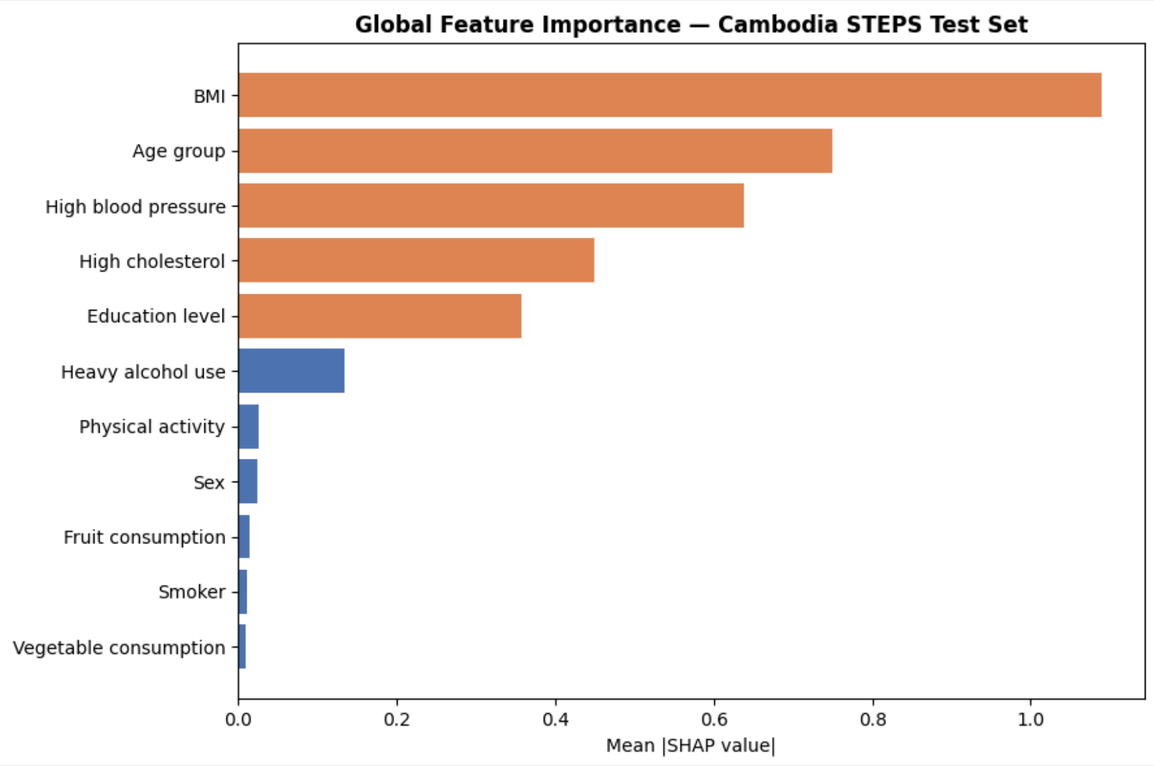
Local Explanation

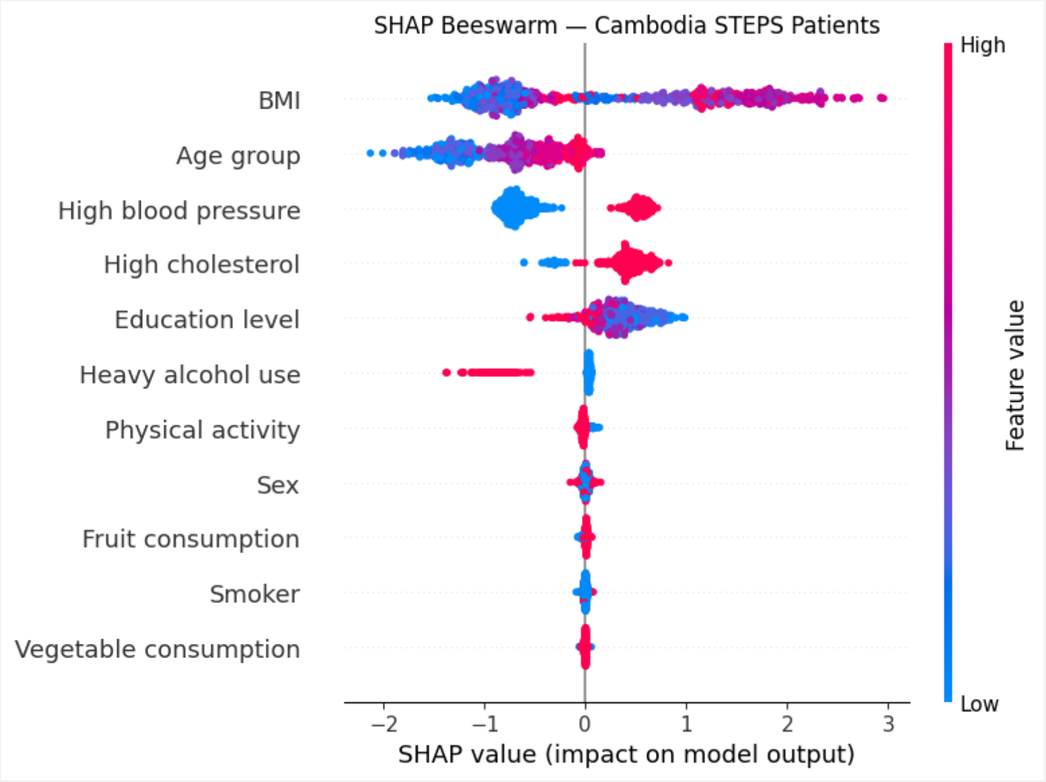
Individual prediction rationale for a single patient using SHAP force plots.

- Explains why the model assigned a specific risk probability to one person
- Shows which features pushed the prediction higher or lower
- SHAP force plot highlights positive and negative contributing factors
- Builds trust by making individual predictions interpretable and actionable

SHAP ANALYSIS

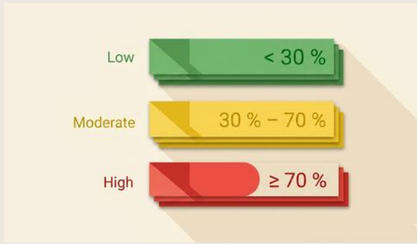
SHAP was used to explain how the final Soft Voting Ensemble model makes diabetes risk predictions





RISK OUTPUT & ACTIONABLE INSIGHTS

Risk Levels



🟢 Low Risk

Probability < 30%

Continue healthy habits and routine check-ups

🟡 Moderate Risk

Probability 30% – 69%

Consult a healthcare provider and monitor key indicators

🔴 High Risk

Probability ≥ 70%

Seek medical advice promptly and take preventive action

Actionable Insights

Actionable Insights:

- Check blood sugar with a healthcare provider
- Maintain a healthy BMI
- Increase physical activity
- Improve diet by reducing sugar and eating more vegetables and fruits
- Monitor blood pressure and cholesterol regularly

Limitation and Future Work

Limitation:

- Cambodia STEPS dataset is small: 873 records
- Data is imbalanced, with only about 14% diabetes cases
- BRFSS and STEPS are from different populations
- No clinical blood test data such as glucose or HbA1c

Future Work:

- Collect more Cambodia health data
- Add clinical features such as glucose and HbA1c
- Improve model performance on local data

CONCLUSION

- Built an explainable AI system for diabetes risk prediction
- Trained models using BRFSS 2017 data
- Tested the final model on Cambodia STEPS
- Selected Soft Voting Ensemble as the final model
- Used SHAP to explain important risk factors
- Generated risk level and actionable recommendations

Thank You